

Core Java Syllabus (Topics Only)

1. Introduction to Java
2. JDK, JRE, JVM
3. Java Program Structure
4. Variables and Data Types
5. Operators
6. Input and Output
7. Control Statements
 - if, if-else
 - switch
 - loops (for, while, do-while)
 - break, continue
8. Arrays
 - 1D Array
 - 2D Array
9. Methods
 - Method Declaration
 - Method Overloading
10. Object-Oriented Programming (OOP)
 - Class and Object
 - Constructor
 - Inheritance
 - Polymorphism
 - Encapsulation
 - Abstraction
11. Packages
12. Interfaces
13. String Handling
 - String
 - StringBuffer
 - StringBuilder

14. Exception Handling

- try-catch
- finally
- throw
- throws
- Custom Exception

15. Wrapper Classes

16. Collections Framework

- ArrayList
- LinkedList
- Vector
- Stack
- HashSet
- TreeSet
- HashMap
- TreeMap
- Iterator

17. Multithreading

- Thread Class
- Runnable Interface
- Synchronization

18. File Handling

- File
- FileReader/FileWriter
- BufferedReader/BufferedWriter

19. Generics

20. Enumerations (Enum)

21. Inner Classes

22. Java I/O Streams

23. Serialization

24. Java 8 Features

- Lambda Expressions
- Functional Interfaces
- Stream API
- Method References

25. JDBC (Database Connectivity)